E-Commerce Platform in Brazil

Customer Segmentation & Lifetime Value

Team 3 | Milestone – 2

1. Our Analytical Focus

In our initial proposal, we identified the 3% repeat purchase rate as the central business problem and proposed a Customer Lifecycle Value-based segmentation approach. During our analysis, we set out to identify which post-purchase experience factors predict repeat purchase. We expected delivery performance and satisfaction to be the primary drivers. Instead, we found that **behavioral engagement signals,** voucher usage and installment adoption are far stronger predictors than service quality. This suggests Olist's retention problem is not about fixing bad experiences, but about **converting passive satisfaction into active engagement**.

2. Data Preparation & Key Decisions

We loaded nine CSV files from the Olist Brazilian e-commerce dataset and performed systematic cleaning to produce analysis-ready tables. The raw dataset contains 99,441 orders spanning September 2016 to August 2018. After filtering to delivered orders and removing 8 records with missing delivery dates, we retained 96,470 orders for analysis.

Data Cleaning Summary

Step	Before	After
Filter to delivered orders	99,441	96,470
Deduplicate reviews	99,224	98,673
Deduplicate geolocation	1,000,163	19,015
Aggregate order_items to order level	112,650	98,666
Aggregate payments to order level	103,886	99,440

Delivery Performance

We computed delivery delay as the difference between actual and estimated delivery date. On average, orders arrive 11.2 days early. However, 8.1% of orders are delivered late, which serves as a key experience signal for our repeat purchase analysis.

Final Outputs

Table	Description
orders_full	Master table: 96,470 orders × 25 columns

rfm_table	Customer-level: 93,349 customers × 18 features
One-time buyers	90,548 (97.0%)
Repeat buyers	2,801 (3.0%)

Key Decisions

Decision	Rationale
Monetary = price only	Freight reflects geography, not purchase intent; kept separate for geo analysis
Common key = customer_unique_id	customer_id can vary across orders for the same person
Repeat buyer = frequency ≥ 2	Binary classification for CLV modeling
Reference date = 2018-08-29	Latest order date in dataset; anchor for recency calculation

Note: Full definitions, verification numbers, and column references are documented in Group3_Shared_Definitions.docx.

3. Descriptive analysis & Initial Findings

In *Figure 1*, it is clearly shown that only 2800 (3%) of 93,349 unique customers came for repeat purchase. However, 90548 (97%) of them were one-time buyers and didn't return for repeat purchases, which is the main issue of the problem we are addressing in this analysis. The huge gap between the two bars shows that retention is where the platform can grow next.

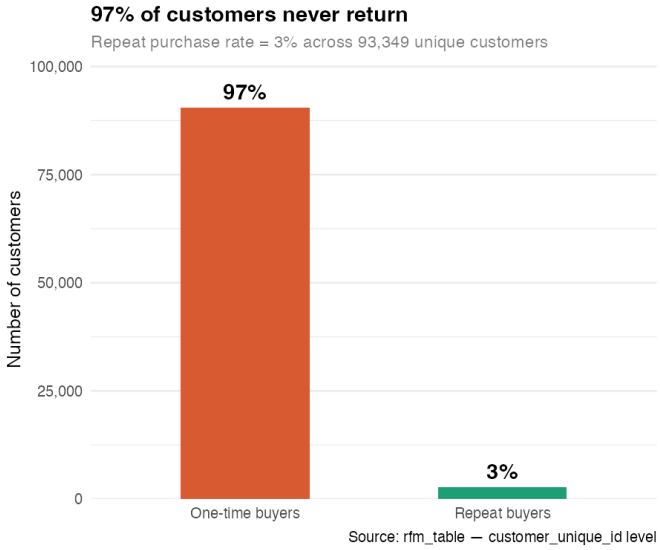


Figure 1: Repeat vs One-time Customers

Figure 2 shows customer spending on a log scale. There is a wide variation in how much customers spend, from under 10 BRL to over BRL 10,000 each. Most customers spend between BRL 50 and BRL 300, with a long right tail of high-value buyers. This variation indicates that not all customers are the same. Therefore, a single retention strategy would not work for all types of buyers. *(Note: BRL stands for **Brazilian Real**, the official currency of Brazil)*

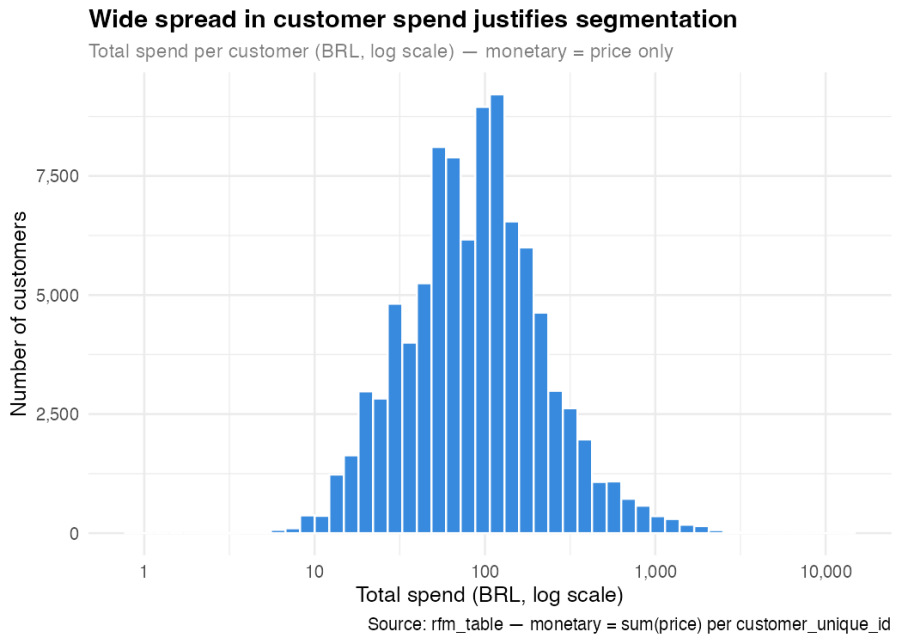


Figure 2: Distribution of Customer Spend

In Figure 3, it is clearly shown that the distribution of delivery delays is left skewed, which suggests that the majority of orders arrive early before the estimated date. The peak of early delivery distribution is approximately 10-15 days. On average, 92% orders arrive early; however, about 8% orders are delivered later than the estimated delivery date. Though it seems to be a smaller number but has an important impact, as the affected customers are more likely to leave negative reviews and might not come back.

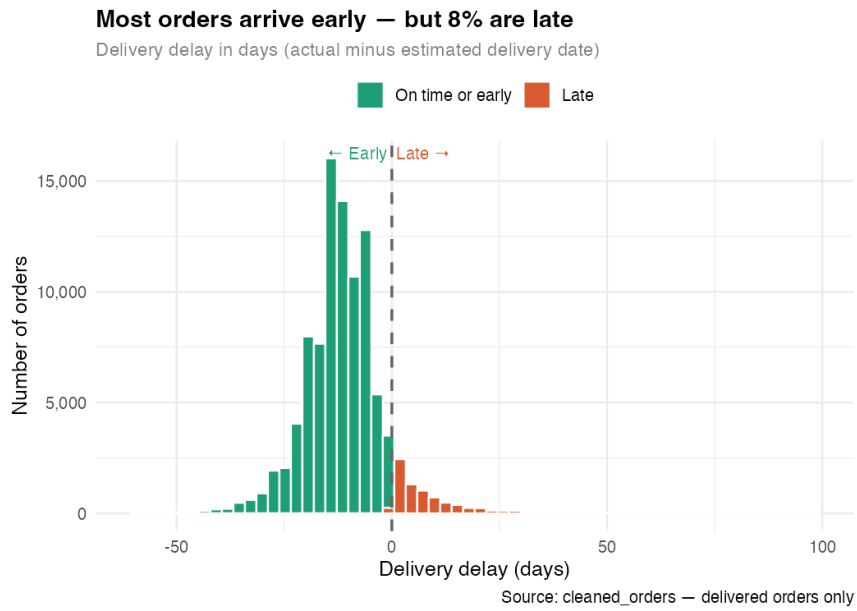


Figure 3: Delivery Delay Distribution

a. CLV & dollar impact

Analytical Approach

To quantify the long-term revenue potential of the platform's customer base, we estimated a Customer Lifetime Value (CLV) score for each of the 93,349 unique customers in the dataset. CLV is calculated by:

CLV = P(repeat) × avg_order_value × expected_frequency

Where:

- **P(repeat)** is the probability that a customer places a second order. Confirmed repeat buyers (frequency ≥ 2) are assigned a probability of 1.0. One-time buyers are assigned the observed platform-wide repeat rate of approximately 3%.
- **avg_order_value** is the mean order spend per customer in BRL, calculated on product price only. Freight costs are excluded, as freight reflects geography rather than willingness of purchase.
- **expected_frequency** is the actual number of orders placed by repeat buyers, and 1 for one-time buyers.

Key Findings

The CLV analysis reveals a sharp and meaningful gap between the platform's two buyer segments. Repeat buyers hold an average CLV of **BRL 260.05**, compared to just **BRL 4.14** for one-time buyers, there is a more than 60x difference. Repeat buyers also place an average of **2.11 orders** within the observation window, compounding their value with each return visit. The platform's current 3% repeat rate means

that most of its customer base (90,500 buyers) fall below this range, representing significant latent revenue that is currently unrealised.

The CLV distribution also confirms that all confirmed repeat buyers fall into the "High" CLV tier. The direct interpretation is that once a customer returns, they become a high-value customer. The challenge, and the opportunity, lies entirely in that first return.

Dollar Impact: Revenue Scenarios

To translate the CLV findings into business terms, we modelled the incremental revenue the platform could generate by improving its repeat purchase. Two scenarios were evaluated:

Incremental revenue = incremental buyers × avg order value × avg repeat frequency

Scenario	Incremental Buyers	Avg Order Value (BRL)	Avg Repeat Frequency	Incremental Revenue (BRL)
3% → 5% repeat rate	1,866	137.51	2.11	**542,532**
3% → 6% repeat rate	2,800	137.51	2.11	**813,876**

A 2% improvement in repeat purchase rate would bring approximately **1,866 additional buyers into a repeat cycle**, generating an estimated **BRL 542,532** in incremental revenue. Doubling that improvement to 6% yields nearly **BRL 814,000**. There is no incremental acquisition cost in either scenario.

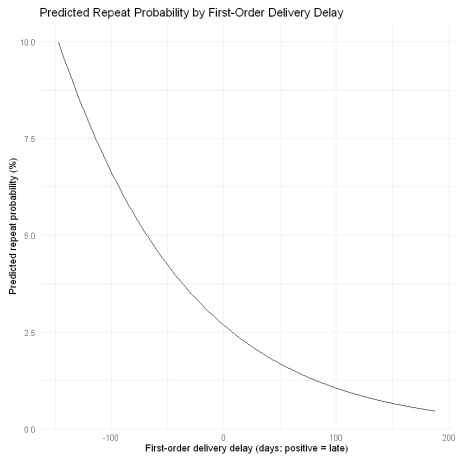
Limitations

The dollar impact calculations apply the average repeat frequency of 2.11 orders. In reality, customers who are successfully converted to repeat buyers may go on to place three, five, or more orders because the most active customer in the dataset placed 15. The scenarios presented here can be a lower-bound estimate of the revenue opportunity.

b. Delivery & experience drivers

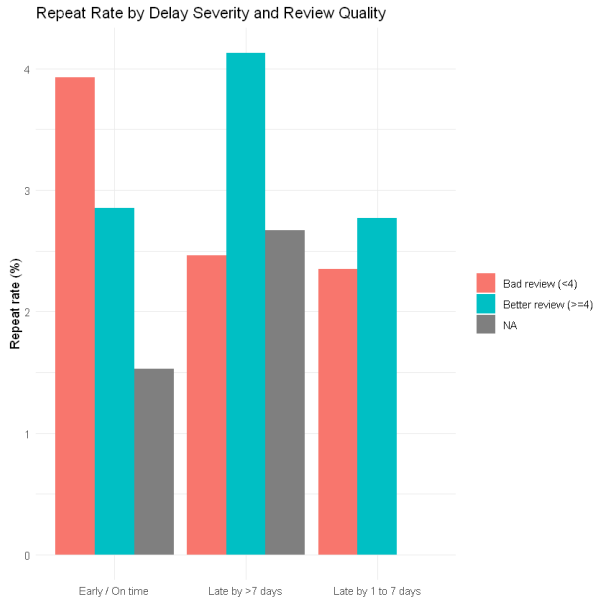
First-Order Delay

We examined whether a customer's initial delivery experience is associated with the likelihood of becoming a repeat buyer. To preserve the timing of the customer journey, we measured delivery delay using the customer's first observed order rather than averaging across all orders. This makes the comparison between one-time and repeat buyers more consistent, since both groups are evaluated on the same initial purchase experience. We found that repeat buyers tended to receive slightly earlier first deliveries and were less likely to experience a late first order than one-time buyers. A logistic regression further confirmed that longer first-order delivery delay is associated with lower repeat likelihood. However, the results did not support a single sharp delay threshold; instead, the effect appears gradual, with repeat probability steadily weakening as delay increases.



Delivery Delay Thresholds

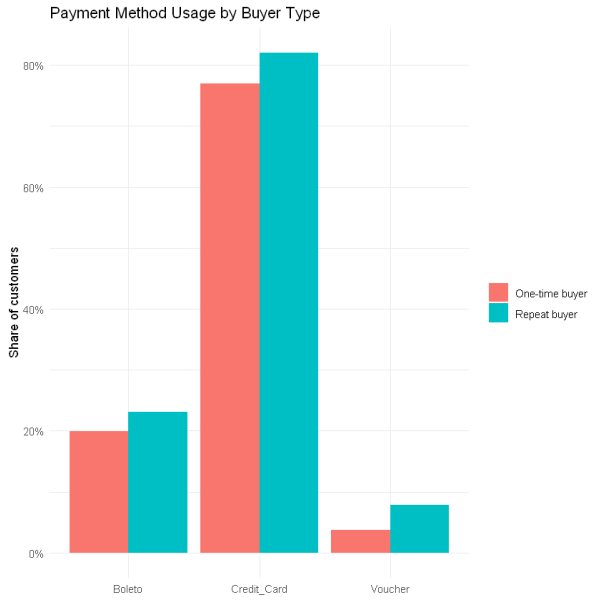
We also explored delivery performance using practical delay bands and looked at whether certain combinations of poor experiences were associated with weaker retention outcomes. This was intended as a supporting, more exploratory view rather than the primary estimate of the delivery-retention relationship. We found that customers whose first orders arrived substantially earlier than expected generally had higher repeat rates, while customers with later deliveries tended to show weaker retention outcomes. At the same time, the banded results did not reveal one exact cutoff where repeat behavior suddenly collapsed. We also used combined views, such as delay severity together with review quality, to identify patterns that may be operationally useful when flagging higher-risk customers.



Experience Signals

We expanded the analysis beyond delivery timing and examined whether other customer experience and transaction signals differed between repeat buyers and one-time buyers. We focused on review

score, installment usage, and payment behavior, since these variables capture post-purchase satisfaction and transaction structure without overlapping with the separate delivery-delay and geographic analyses. We found that repeat buyers tended to have slightly higher average review scores, use more installments, and show different payment-method usage patterns than one-time buyers. A logistic regression indicated that these variables are positively associated with repeat-buyer status when examined together. Overall, the results suggest that customer satisfaction and transaction behavior help distinguish customers who return from those who do not, although payment variables should be interpreted primarily as behavioral markers rather than direct causal drivers of retention.

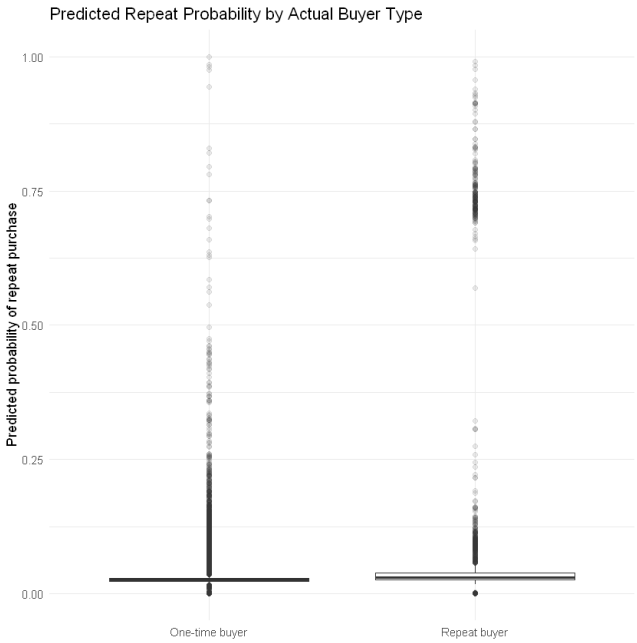


c. Binary classifier

We used logistic regression with 8 predictor variables capturing first-order experience signals. Importantly, we excluded recency and frequency from the model to avoid data leakage, as these variables directly define the target (repeat buyer = frequency ≥ 2). The model was trained on a 70/30 train-test split (64,921 training / 27,824 test) with seed 520 for reproducibility.

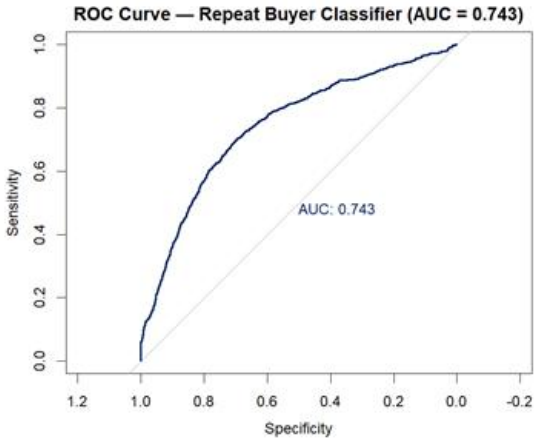
Repeat vs One Time Classifier

We converted the earlier descriptive findings into an actionable customer-level prediction model. Using customer spending, review score, installment usage, payment-method indicators, and first-order delivery delay, we estimated a probability of repeat purchase for each customer. Because repeat buyers are rare in the dataset, we did not treat the model as a strict 0.50-threshold classifier. Instead, we used it as a ranking and targeting tool. We found that higher spending, more installments, and certain payment behaviors are associated with higher repeat likelihood, while longer first-order delivery delay is associated with lower repeat likelihood. In practice, the resulting P(repeat) scores are most useful when evaluated at lower operational thresholds, such as 5% or 10%, and when combined with CLV tiers to identify customers who are both more likely to return and more valuable to the business.



Model Performance

The model achieved an **AUC of 0.743**, indicating fair discriminative ability. This means the model correctly ranks a random repeat buyer above a random one-time buyer 74.3% of the time. Given the highly imbalanced target (only 3% repeat rate) and deliberate exclusion of leakage-prone features, this represents solid predictive performance.

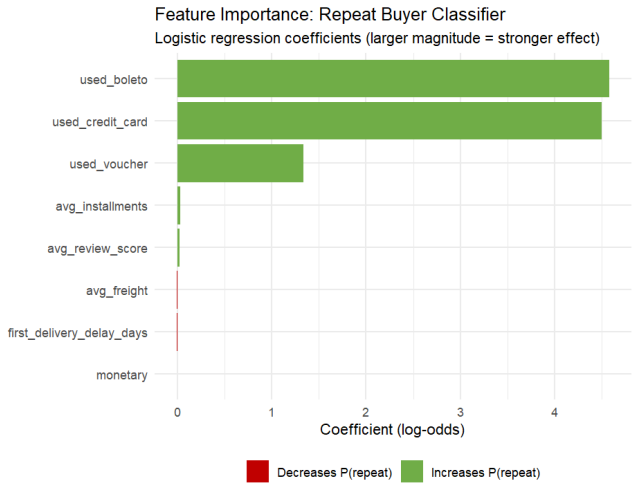


Feature Importance Results

The below table presents the ranked feature importance from the logistic regression model. Payment method variables (credit card, boleto) showed artificially high coefficients due to the reference group issue (nearly all customers use at least one payment method) and are excluded from interpretation.

Rank	Feature	Effect	Odds Ratio	Interpretation	Significance
1	Voucher usage	↑ Increases	3.81x	+281% odds	***
2	Installments	↑ Increases	1.03x	+2.6% per unit	**
3	Monetary	↑ Increases	1.001x	+0.14% per BRL	***
4	Freight cost	↓ Decreases	0.99x	-1.1% per BRL	***
5	First-order delay	↓ Decreases	0.99x	-0.76% per day	**
6	Review score	—	1.02x	Not significant	

*Note: *** p < 0.001, ** p < 0.01*



Key Findings

Voucher usage is the strongest predictor of repeat purchase. Customers who used vouchers on their first order are 3.8x more likely to return. This suggests that loyalty program engagement signals future purchase intent, or that voucher-driven customers are already predisposed to repeat behavior.

Installment usage increases repeat probability. Each additional installment is associated with a 2.6% increase in odds of repeat purchase. Customers comfortable with installment payments may be more financially engaged with the platform.

Higher freight costs reduce repeat probability. Each additional BRL in freight is associated with a 1.1% decrease in odds. This is particularly relevant for Olist's geographically diverse customer base, where remote customers face higher shipping costs.

First-order delivery delay matters. Each day of delay reduces repeat odds by 0.76%. A 14-day late delivery reduces repeat odds by approximately 10%. First impressions matter—customers who experience delivery problems on their first order are less likely to return.

Review scores is not a significant predictor. While repeat buyers have slightly higher average review scores (4.20 vs 4.15), this difference is not statistically significant in the multivariate model. Satisfaction alone does not drive repeat behavior.

Customer Scoring

We applied the model to score all 92,745 customers with complete data. Below table shows the distribution of predicted repeat probabilities.

P(repeat) Band	Customers	Share
Very high (≥20%)	403	0.4%
High (10-20%)	799	0.9%
Medium (5-10%)	3,354	3.6%
Low (2-5%)	78,981	85.2%
Very low (<2%)	9,208	9.9%

High-CLV customers have significantly higher average P(repeat) scores (4.47%) compared to Low-CLV customers (2.44%), suggesting that targeting high-CLV customers with retention campaigns would yield the best ROI.

Business Implications

Based on these findings, we recommend the following strategies to improve repeat purchase rates:

1. Expand voucher programs. Voucher usage is the strongest signal of repeat intent. Consider offering first-purchase vouchers or loyalty rewards to drive engagement.
2. Promote installment options. Customers who use installments are more likely to return. Make installment options more visible at checkout.
3. Prioritize first-order delivery. First impressions matter. Consider expedited shipping or delivery guarantees for new customers.
4. Subsidize freight for high-value customers. Higher freight costs reduce repeat probability. For high-CLV customers in remote regions, consider freight subsidies as a retention investment.

4. Next Steps

Regional CLV Mapping: CLV scores will be joined to the geolocation table to map high-value buyer concentration by Brazilian state and city, identifying the top regions where acquisition investment is most likely to generate buyers with the highest long-term return.
Regional Segment Sizing: Each region will be sized to show how many high-CLV buyers are in each one, so the platform understands not just where the opportunity exists but how large it is in each market.

Category Concentration: The analysis will identify which product categories attract more high-CLV buyers by cross-referencing repeat purchase rate by category, giving the platform a view into which categories attract buyers to stay.
Tenure & Time-Between-Orders: The time between a customer's first and second order will be examined across categories to inform how quickly the platform should act after an initial purchase.
Cohort Analysis: Buyers will be grouped by acquisition period to assess whether retention patterns shifted as the platform grew, telling us whether the platform's ability to generate repeat buyers has improved or deteriorated over time.
Seller-Level Analysis: Seller-level patterns will be examined to identify whether certain sellers drive more repeat purchase behaviour.
Segmentation Matrix: All findings across CLV, experience signals, geographic mapping, and category analysis will be compiled into a decision framework with columns for segment name, estimated CLV, key experience drivers, and recommended intervention. For example, high-CLV one-time buyers who experienced delivery delays would receive a recovery offer.